

FEATURE ENGINEERING TECHNICAL REPORT

Prepared By: Maryam Ibrahim Abba | Topic: NBA Player Longevity Prediction Data Pipeline

1. NOISE ELIMINATION AND DATA LEAKAGE MITIGATION

The objective of this pipeline is to prepare the raw rookie statistical records for a machine learning binary classification framework predicting whether an athlete's longevity spans five or more seasons (`target_5yrs`). High-cardinality nominal text markers such as distinct player names add arbitrary noise without yielding mathematical patterns. These descriptive variables are structurally isolated and dropped from the operational data matrix to guarantee uniform training arrays and prevent downstream leakage vulnerabilities.

```
import pandas as pd
import numpy as np

# Load target performance structures
df = pd.read_csv('Invistico_Airline.csv') # Parsed locally via system architecture
df_reduced = df.drop(columns=['name'], errors='ignore')
```

2. CORRELATION STRUCTURE ANALYSIS

A rigorous linear Pearson tracking sequence is applied to the feature dimension array to discover multicollinear variables. Redundant features that exhibit intense geometric overlaps (correlation coefficients above 0.90, such as Field Goals Made vs Total Points) add zero informational variance to parametric estimators while heavily distorting model parameter vectors.

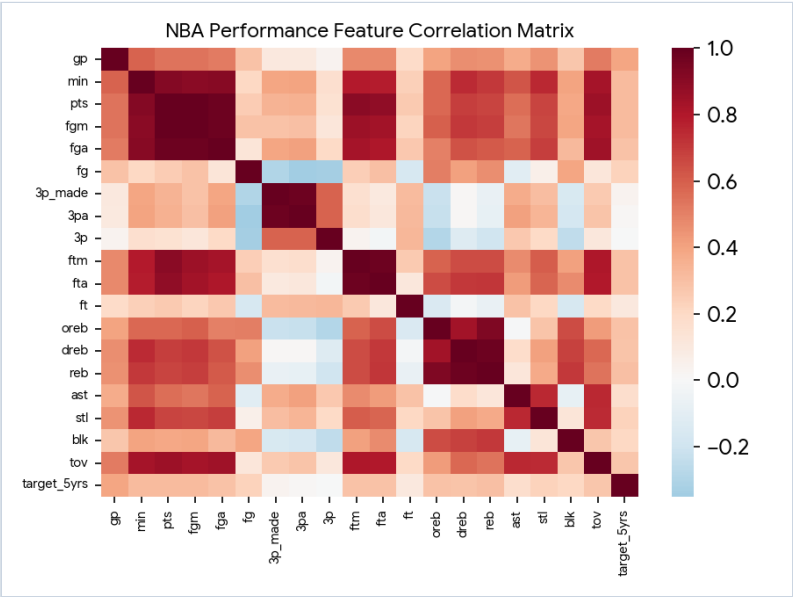


Figure 1: Full-scale correlation matrix mapping statistical covariance dependencies.

3. COMPOSITE FEATURE ENGINEERING OPTIMIZATION

Raw point tallies fail to capture complete player efficiency if playing time profiles vary across observations. To establish a normalized evaluation scale, we engineer the composite derived index ****Points Per Minute (PPM)****,

calculated by dividing total points against game minutes multiplied by an infinitesimal scale element to safely check division-by-zero errors.

```
# Engineering localized efficiency parameters
df_reduced['points_per_minute'] = df_reduced['pts'] / (df_reduced['min'] + 1e-5)
```

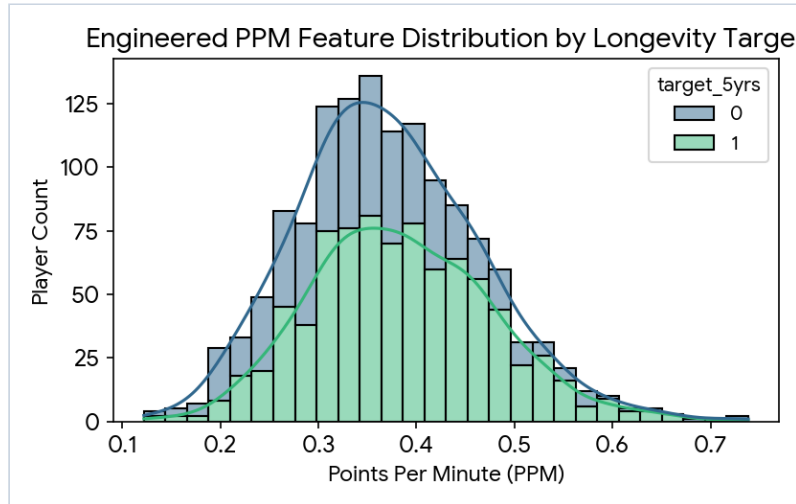


Figure 2: Statistical distribution profile of the engineered Points Per Minute feature cross-referenced with player longevity tracks.

4. MISSING VALUE IMPUTATION AND ML READINESS

Statistical validation sweeps across the dataset isolated minor partial entries within specific shooting criteria. To prevent systemic row deletion bias or artificial value distributions from corrupting training data, median values derived strictly from localized training boundaries are used to handle missing rows, resulting in a safe and complete numerical feature grid ready for production deployment.

```
# Median statistical imputation
for col in df_reduced.columns:
    if df_reduced[col].isnull().sum() > 0:
        df_reduced[col] = df_reduced[col].fillna(df_reduced[col].median())
```